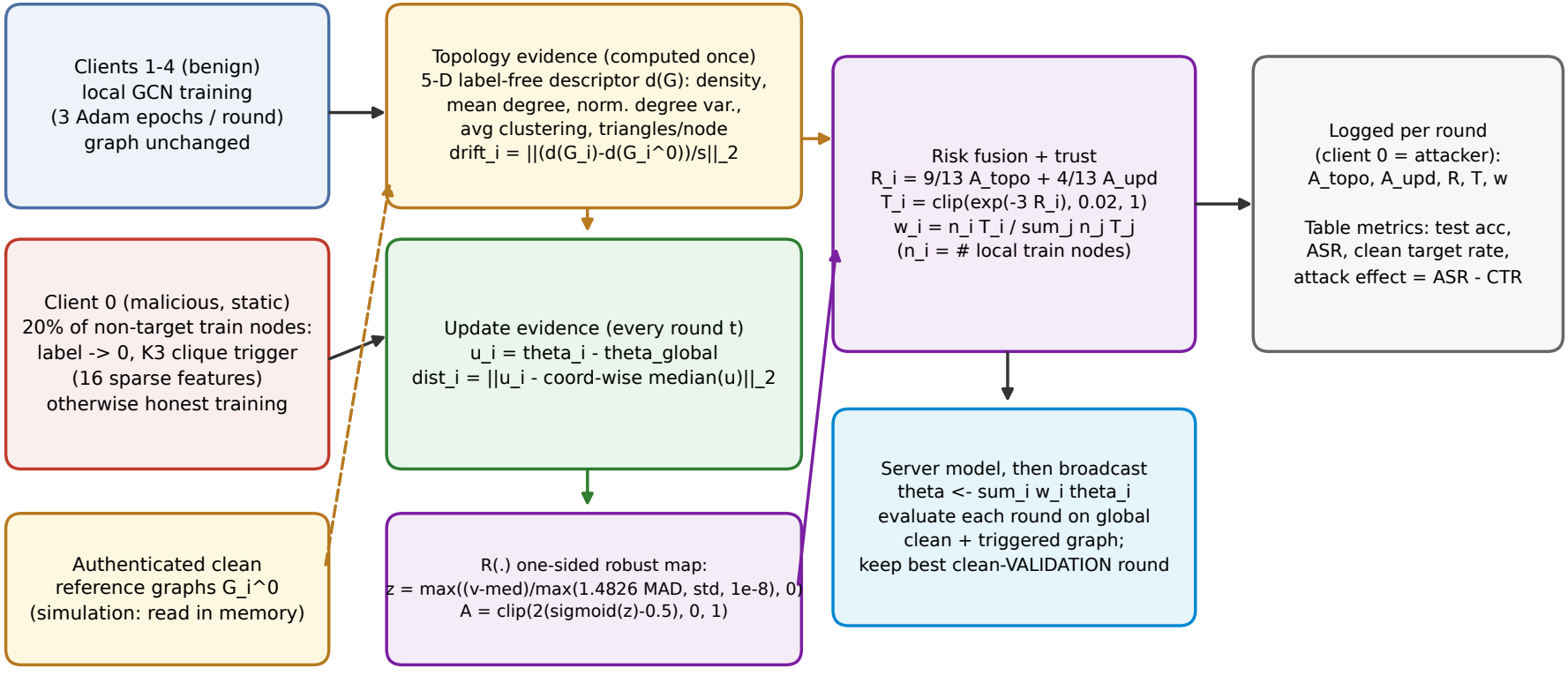
Simulation assumptions: K=5, M=1 attacker, IID class-stratified partition, server reads clean AND current client graphs; not a privacy-preserving deployment
Clients 1-4 (benign)
local GCN training
(3 Adam epochs / round)
graph unchanged
Client 0 (malicious, static)
20% of non-target train nodes:
label -> 0, K3 clique trigger
(16 sparse features)
otherwise honest training
Authenticated clean
reference graphs G_i^0
(simulation: read in memory)
Topology evidence (computed once)
5-D label-free descriptor d(G): density,
mean degree, norm. degree var.,
avg clustering, triangles/node
drift_i = ||(d(G_i)-d(G_i^0))/s||_2
Update evidence (every round t)
u_i = theta_i - theta_global
dist_i = ||u_i - coord-wise median(u)||_2
R(.) one-sided robust map:
z = max((v-med)/max(1.4826 MAD, std, 1e-8), 0)
A = clip(2(sigmoid(z)-0.5), 0, 1)
Risk fusion + trust
R_i = 9/13 A_topo + 4/13 A_upd
T_i = clip(exp(-3 R_i), 0.02, 1)
w_i = n_i T_i / sum_j n_j T_j
(n_i = # local train nodes)
Server model, then broadcast
theta <- sum_i w_i theta_i
evaluate each round on global
clean + triggered graph;
keep best clean-VALIDATION round
Logged per round
(client 0 = attacker):
A_topo, A_upd, R, T, w
Table metrics: test acc,
ASR, clean target rate,
attack effect = ASR - CTR